

New $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ Solid Solution: Structure, Microstructure, and Li^+ Conductivity

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The stoichiometry range, crystal chemistry, ionic conductivity, and electrochemical window of the $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ system with a perovskite-related structure have been studied. The range of existence of the solid solution is ($0.06 \leq x \leq 0.3$). Powder X-ray diffraction and transmission electron microscopy results show that these oxides have a unit cell multiple of the perovskite cell with dimensions $a \approx \sqrt{2} a_p$, $b \approx \sqrt{2} a_p$, and $c \approx 2a_p$. The ionic conductivity of the material and its dependence on composition and temperature have been examined by complex impedance spectroscopy. Lithium ion conductivity increases with increasing lithium content up to the value $7.12 \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at room temperature, corresponding to $x = 0.25$. Electrochemical experiments indicate that these materials can be used as solid electrolytes in secondary batteries having Li metal as the anode down to 1.75 V.

Introduction

Lithium batteries are among the most promising power sources for the future due to their high energy density and good cycling life. In particular, all-solid-state lithium batteries allow fabrication of compact and leak proof batteries of different shapes and sizes. Therefore, research on materials as potential electrolytes in solid-state lithium batteries is an interesting area of solid-state science.

The compounds of general formula $\text{La}_{2/3-x}\text{Li}_x\text{TiO}_3$ have the highest lithium conductivity at room temperature ($\sigma_{\text{bulk}} \approx 1 \times 10^{-3} \text{ S}\cdot\text{cm}^{-1}$) reported to date.^{1–5} Pure phases are found for a range of compositions corresponding to $0.036 \leq x \leq 0.14$,^{3,6} and the oxides have a crystal structure related to the perovskite type (ABO_3) with A-cation deficiency. Other systems such as $\text{RE}_{2/3-x}\text{Li}_x\text{TiO}_3$ ($\text{RE} = \text{Pr}, \text{Nd}, \text{Sm}$) have been studied,^{3,6–8} but in all of them it has so far been found that the lithium conductivity is lower than that in the La system. No significant changes are seen in the crystal structure apart from the change in size of the unit cell. It then seems that the decrease of the RE–O distances (as a consequence of the smaller ionic radii of Pr, Nd, and Sm compared to La) reduces the bottleneck size, dis-

turbing the Li^+ ion migration via A-size vacancies. Moreover, in situ experiments concerning the ionic conductivity dependence on mechanical pressure for $\text{La}_{0.52}\text{Li}_{0.35}\text{TiO}_{2.96}$ indicate that the conductivity decreases when the pressure increases,⁹ i.e., the pressure effect on the ionic conductivity seems to be equivalent to the A-size substitution effect. Therefore, increasing the A–O distances could eventually improve the lithium conductivity of these materials.

The perovskite structure tolerates total or partial substitution of the A and B cations by ions with different valence states. Sr^{2+} and Ba^{2+} are bigger cations than La^{3+} , so it is interesting to study the influence on the ionic conductivity of the substitution of La^{3+} in the $\text{La}_{2/3-x}\text{Li}_x\text{TiO}_3$ system by these divalent cations. In this sense, an increase of the bulk ionic conductivity ($\sigma_{\text{bulk}} \approx 1.5 \times 10^{-3} \text{ S}\cdot\text{cm}^{-1}$) has been reported in the $(\text{La}_{1/2}\text{Li}_{1/2})_{1-x}\text{M}_x\text{TiO}_3$ ($\text{M} = \text{Sr}, \text{Ba}$) oxides for $\text{M} = \text{Sr}$ and $x = 0.05$.⁴ More recently, a slight improvement of the lithium conductivity ($\sigma_{300\text{K}} \approx 1.12 \times 10^{-3} \text{ S}\cdot\text{cm}^{-1}$) has been demonstrated in $\text{La}_{0.57-2x/3}\text{Sr}_x\text{Li}_{0.3}\text{TiO}_3$ for $x = 0.08$.¹⁰ In this paper, we report the simultaneous lanthanum substitution by strontium and lithium addition in the system $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$. This system allows us to study the ionic conductivity dependence on both the size of the crystal unit cell and the carrier concentration. We present here our data concerning the stoichiometry range, crystal chemistry, and lithium ion conductivity, as well as the electrochemical window versus Li, for the $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ solid solution.

Experimental Section

Different compositions of the $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ system were prepared from stoichiometric amounts of Li_2CO_3 (Merck 99%),

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Table 1. Molar Ratio of the Metal Ions for Three Samples of the System $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$

nominal composition		$\text{La}_{0.61}\text{Sr}_{0.06}\text{Li}_{0.06}\text{TiO}_3$	$\text{La}_{0.52}\text{Sr}_{0.15}\text{Li}_{0.15}\text{TiO}_3$	$\text{La}_{0.42}\text{Sr}_{0.25}\text{Li}_{0.25}\text{TiO}_3$
ICP	titanium	1 ± 0.01	1 ± 0.01	1 ± 0.01
	lithium	0.05 ± 0.02	0.17 ± 0.02	0.25 ± 0.02
	strontium	0.04 ± 0.02	0.14 ± 0.02	0.22 ± 0.02
XRFS	titanium	1 ± 0.01	1 ± 0.01	1 ± 0.01
	lanthanum	0.61 ± 0.02	0.53 ± 0.02	0.42 ± 0.02
	strontium	0.06 ± 0.02	0.15 ± 0.02	0.24 ± 0.02

La_2O_3 (Aldrich 99.999%), TiO_2 (Aldrich 99.99%), and SrCO_3 (Aldrich 99.9%). La_2O_3 was heated overnight at 1000 °C and TiO_2 was heated at 700 °C prior to weighing. The mixtures were ground and then heated in platinum boats for 6 h for decarbonation. Afterward, the samples were reground, pelleted, covered with powder of the sample of the same composition to prevent lithia loss, and fired at 1100 °C for 12 h, followed by further grinding, repelleting, and refiring for another 12 h at two different temperatures depending on the composition: at 1170 °C for the samples with $0.15 \leq x \leq 0.3$ and at 1275 °C for the samples with $0.06 \leq x < 0.15$. To study the evolution of the crystal structure with temperature, small amounts of sample were wrapped in Pt foil envelopes, annealed isothermally at different temperatures, and quenched to room temperature on a brass plate.

The molar ratio of the metals in the compounds was determined by inductively coupled plasma mass spectroscopy (ICP-MS) and X-ray fluorescence spectroscopy (XRFS). For ICP-MS we used a Perkin-Elmer Elan 6000 with autosampler AS. To perform the ICP analysis, a solution protocol was optimized. Samples were treated with a mixture of nitric, hydrochloric, and hydrofluoric acids in a 3:2:1 molar ratio at 150 °C in a CEM microwave furnace digester (MDS-2000 model) with temperature and pressure sensors. Because of La^{3+} precipitation during the dilution with the acid solution, XRFS was used to analyze this cation using a Bruker S4 Explorer.

Crystalline phase identification and determination of the lattice parameters were carried out by powder X-ray diffraction (XRD) using a Philips X'PERT diffractometer with $\text{Cu K}\alpha 1$ radiation, a curved Cu monochromator, and the Philips X'PERT PEAPD software. The patterns were taken at step mode with a step size equal to 0.02 (2θ degrees) and time per step equal to 10 s.

The microstructure of the materials was characterized by selected area electron diffraction (SAED) and high-resolution transmission electron microscopy (HRTEM). Samples were ground in *n*-butyl alcohol and ultrasonically dispersed, and drops of the resulting suspension were deposited in carbon-coated grids. SAED studies were performed with an electron microscope JEOL 2000 FX (double tilt $\pm 45^\circ$) working at 200 kV, and HRTEM studies were performed with an electron microscope JEOL 4000 FX (double tilt $\pm 25^\circ$) working at 400 kV.

The electrical behavior of the materials has been studied by an impedance/gain phase analyzer Solartron 1255A with dielectric interface 1296. Pellets of about 13 mm diameter and 2 mm thickness were prepared by pressing the powder samples and sintering them at 1200 °C. Electrodes were made by coating opposite pellet faces with platinum paste and heating to 850 °C. Measurements were carried out in air on a heating cycle from room temperature to 100 °C and in a frequency range of 1×10^{-3} to 6.5×10^6 Hz.

The electrochemical window of the materials was examined by potentiostatic and galvanostatic discharge and charge using Swagelok cells and a multichannel potentiostatic/galvanostatic MacPile system.¹¹ Li foil was used as counter electrode, and 1 M solution of LiPF_6 in ethylene carbonate (EC) and dimethylene carbonate (DMC) was used as electrolyte. The working electrode was a mixture of the samples, carbon black, and binder in the ratio 87:8.7:4.3 (in wt %). Cells were assembled

in an argon-filled glovebox. Cyclic voltametries were carried out within the potential range 0–3 V vs Li/Li^+ at the scanning rate of 20 mV/2 h. Cells were galvanostatically discharged and charged using a current density of 30 $\mu\text{A}/\text{cm}^2$ between 3 and 1 V.

Results and Discussion

Stoichiometric Range and Crystal Chemistry of the Solid Solutions $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$. Single phase oxides are found over the composition range $0.06 \leq x \leq 0.3$. Compositions out of this range were multiphase. Comparing this composition range with that corresponding to other systems, such as $\text{RE}_{2/3-x}\text{Li}_x\text{TiO}_3$ ($\text{RE} = \text{La, Nd, Pr}$)^{3,6,8} and $\text{La}_{2/3}\text{Li}_x\text{Ti}_{1-x}\text{Al}_x\text{O}_3$,¹² led to the conclusion that this kind of solid solution seems to admit a number of vacancies within the A positions up to about 27% and down to about 4%, under the preparation conditions described. Table 1 shows the chemical composition analysis results of three samples with different x values. Owing to the good agreement between analytical and nominal values, nominal compositions will be used in the following.

Figure 1a shows the powder X-ray diffraction patterns of four samples quenched from the temperature of synthesis. The oxides with $x = 0.3, 0.2$ have a perovskite-type structure. Superlattice reflections with respect to the ideal cubic perovskite structure are observed in the patterns corresponding to $x = 0.06, 0.1$. These reflections are broad and have low intensity in the patterns for $x = 0.1$ but are well defined when the sample is annealed at a temperature lower than that during the synthesis. Figure 1b shows the powder XRD patterns of four oxides annealed at 1000 °C and quenched from this temperature. We can see that the intensity of the superlattice reflections increases when annealing the oxides with $0.06 \leq x \leq 0.15$. However, no extra reflections appear at any annealing temperature in the patterns of the oxides with $0.15 \leq x \leq 0.3$. The superlattice reflections double the c_p lattice parameter (p refers to the cubic perovskite unit cell). Strictly speaking, perovskites do not need to define c_p as they have cubic symmetry and only a_p is used; however, it is convenient for the discussion to choose a direction for the doubling of the structure: this is usually taken as c .

A similar dependence of the crystal structure on annealing temperature and composition of the materials has been found in the system $\text{RE}_{2/3-x}\text{Li}_x\text{TiO}_3$ ($\text{RE} = \text{La, Nd, Pr}$)^{3,6,8} and $\text{La}_{2/3}\text{Li}_x\text{Ti}_{1-x}\text{Al}_x\text{O}_3$.¹²

The microstructure of the oxides has been studied by SAED and HRTEM. Figure 2 (a, b, and c) shows three representative patterns of the oxide with $x = 0.06$ quenched from the synthesis temperature. The Bragg

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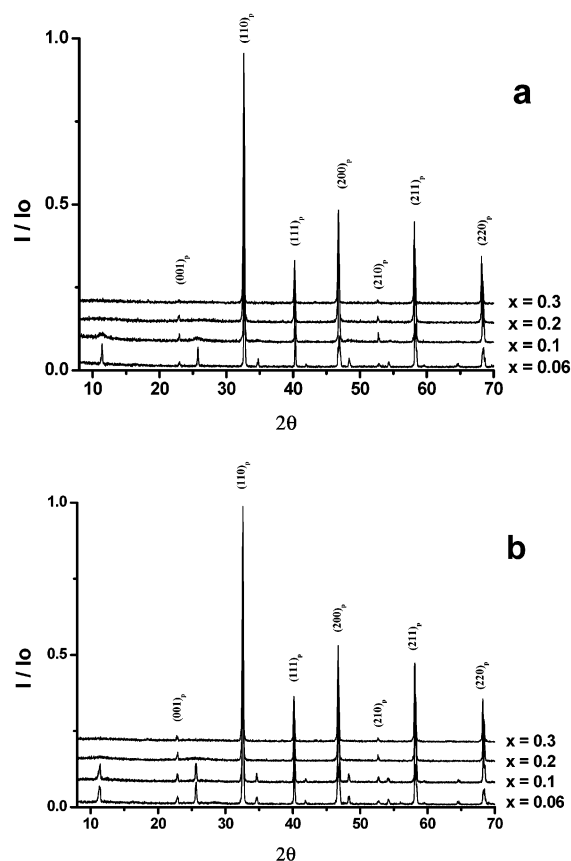


Figure 1. Powder XRD patterns of (a) $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.06, 0.1, 0.2$, and 0.3) quenched from the temperature of synthesis; (b) $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.06, 0.1, 0.2$, and 0.3) annealed at 1000°C and quenched from this temperature. Miller indices of the cubic perovskite are indicated in the patterns corresponding to $x = 0.3$.

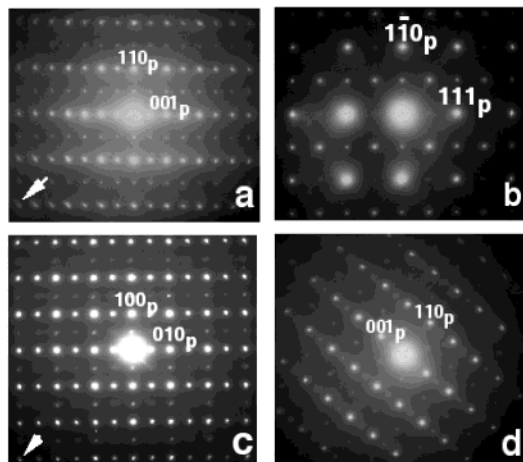


Figure 2. SAED patterns of $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.06$) quenched from the synthesis temperature along the $[110]_p$ (a); $[112]_p$ (b); and $[001]_p$ (c) zone axes. SAED pattern of $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.2$) annealed at 1000°C and quenched from this temperature along the $[110]_p$ zone axes (d).

reflections have been indexed according to the ideal perovskite cell. Extra superlattice reflections are observed: $(0\ 0\ 1/2)_p$ reflections which double the c -axis in the pattern of the $[110]_p$ zone axis; $(h/2\ k/2\ 1/2)_p$ reflections in the three patterns and $(h/2\ 0\ 0)_p$ and $(0\ k/2\ 0)_p$ reflections, which apparently double the other two a_p and b_p lattice parameters, in the pattern of the $[001]_p$ zone axis.

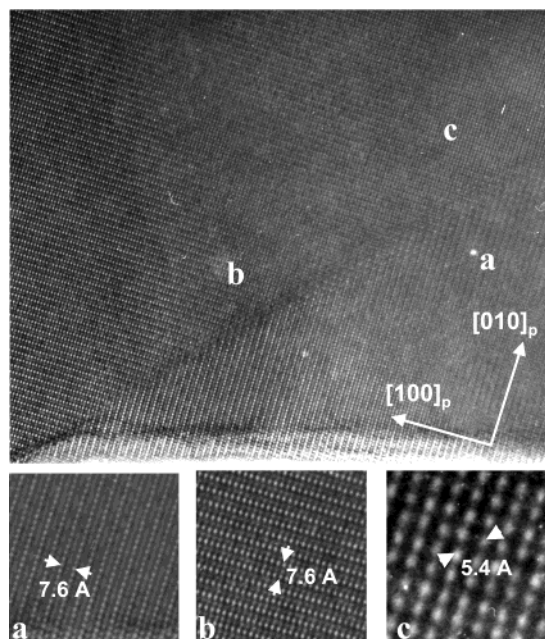


Figure 3. HRTEM image corresponding to the $[001]_p$ zone axes of $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.06$) quenched from the synthesis temperature and magnifications of the three domains (a, b, c) observed in the image of the crystal.

According to all these intensity maxima, the unit cell of the materials is either $2a_p \times 2a_p \times 2a_p$ or $\sqrt{2}a_p \times \sqrt{2}a_p \times 2a_p$; but, in this second case, the crystals should be twinned due to the different orientations of the c -axis. Electron microscopy was used to clarify which of the two possibilities correspond to our materials. Figure 3 shows the HRTEM image of the $[001]_p$ zone axis of the sample with $x = 0.06$ quenched from the synthesis temperature. The crystal has three different domains: one domain with image difference contrasts corresponding to $2a_p$ periodicity along the $[001]_p$ direction; another domain showing the $2a_p$ periodicity along $[010]_p$ direction; and the third domain with $\sqrt{2}a_p$ periodicity along the $[110]_p$ direction. The image confirms that the unit cell of the material is $\sqrt{2}a_p \times \sqrt{2}a_p \times 2a_p$ and that the crystal is formed by three sets of domains with different orientation of the c -axis within each of them. Coming back to the SAED patterns, it is important to comment that far away from the origin, splitting of some reflections into two can be seen (see arrows in the patterns of the $[110]_p$ and $[001]_p$ zone axes of Figure 2). This is due to the slightly different size of the a and b lattice parameters and the twinning of the crystal. The observed symmetry of the crystal is then orthorhombic.

All the samples have similar SAED patterns with no obvious dependence on either annealing temperature or composition, indicating that the crystal structure of all these materials is based in the unit cell $\sqrt{2}a_p \times \sqrt{2}a_p \times 2a_p$. The main difference found is that the $(0\ 0\ 1/2)_p$ reflections are very weak in the patterns of the materials with $0.15 < x \leq 0.3$ (notice that these reflections were absent in the XRD patterns) and also that the $(h/2\ k/2\ 1/2)_p$ reflections with l even are extremely weak, almost invisible (see Figure 2d). Yet, the microstructure of the compounds within the $0.15 < x \leq 0.3$ regime is somewhat different. Figure 4 shows the HRTEM image of the $[001]_p$ zone axis of the sample corresponding to $x = 0.2$ annealed at 1000°C and quenched from this

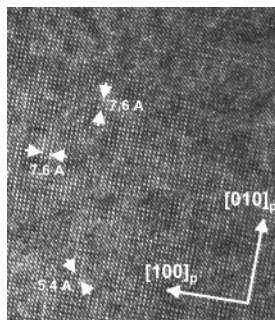


Figure 4. HRTEM image corresponding to the $[001]_p$ zone axes of $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.2$) annealed at 1000 °C and quenched from this temperature.

Table 2. Lattice Parameters of $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$

x	a (Å)	b (Å)	c (Å)	V (Å ³)
0.06	5.477 ± 0.006	5.474 ± 0.003	7.767 ± 0.004	232.95 ± 0.02
0.1	5.481 ± 0.003	5.476 ± 0.002	7.768 ± 0.004	233.13 ± 0.01
0.15	5.489 ± 0.002	5.485 ± 0.004	7.769 ± 0.003	233.87 ± 0.01
0.2	5.493 ± 0.004	5.490 ± 0.004	7.771 ± 0.003	234.41 ± 0.02
0.25	5.495 ± 0.003	5.491 ± 0.001	7.772 ± 0.004	234.52 ± 0.02
0.275	5.497 ± 0.003	5.493 ± 0.001	7.774 ± 0.003	234.73 ± 0.01
0.3	5.499 ± 0.003	5.494 ± 0.001	7.775 ± 0.001	234.88 ± 0.01

temperature. The $2a_p$ periodicity is observed in small regions of the crystal of just two or three times the size of the unit cell dimensions. These crystals are not formed by large domains but by very small ones showing the three different orientations of the c -axis.

The most common distortion of the perovskite structure consists of the tilting of the BO_6 octahedra, leading to the $\sqrt{2} a_p \times \sqrt{2} a_p \times 2a_p$ unit cell. Tilting of the TiO_6 octahedra must occur in our materials to accommodate the size of the A cations (La^{3+} , Li^+ , and Sr^{2+}), to attain the appropriate A–O distances. Moreover, an ordering of the A cations could also exist. It has, in fact, been reported in other compounds with perovskite-related structure that the presence of the $(0\ 0\ 1/2)_p$ reflection in the powder X-ray diffraction patterns indicates an A-cation ordering perpendicular to the c -axis.¹³ However, the intensity of this reflection seems to depend on the size of the diffracting crystal domains. Our results show that the samples giving X-ray diffraction patterns with strong $(0\ 0\ 1/2)_p$ reflection are characterized by relatively big domains in the crystals (Figure 3) whereas very broad reflections are observed in the case of a microdomain texture (Figure 4).

The powder X-ray diffraction patterns of the oxides have, consequently, been indexed using the orthorhombic unit cell $\sqrt{2} a_p \times \sqrt{2} a_p \times 2a_p$ so as to calculate the variation of the cell size with La^{3+} substitution. Table 2 shows the lattice parameters of some oxides with different composition. The lattice parameters increase when x increases, as it is expected of the substitution of La^{3+} by Sr^{2+} and Li^+ (Sr^{2+} has a larger ionic radius than La^{3+} at equal coordination).

Electrical Properties of the $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($0.06 \leq x \leq 0.3$) System. All the materials of the solid solution have a similar electrical behavior. We will describe in the following the data corresponding to a sample with $x = 0.25$.

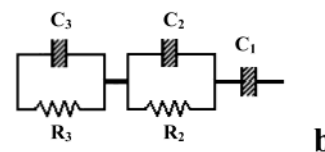
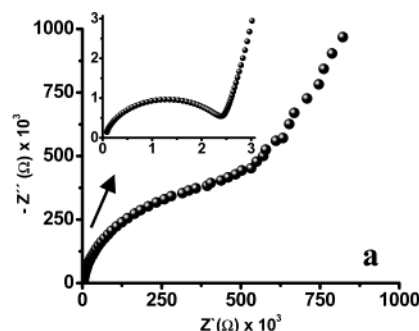


Figure 5. (a) Complex impedance diagram at 22 °C of $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.25$); (b) equivalent circuit to model the data.

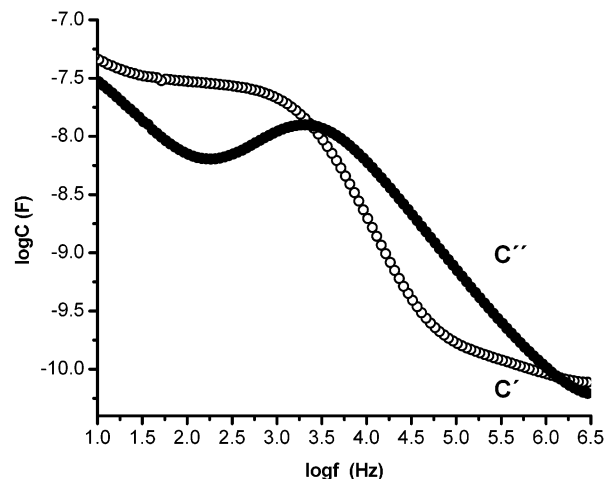


Figure 6. $\log C'(\omega)$ and $\log C''(\omega)$ vs $\log$ of frequency at 22 °C for $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.25$).

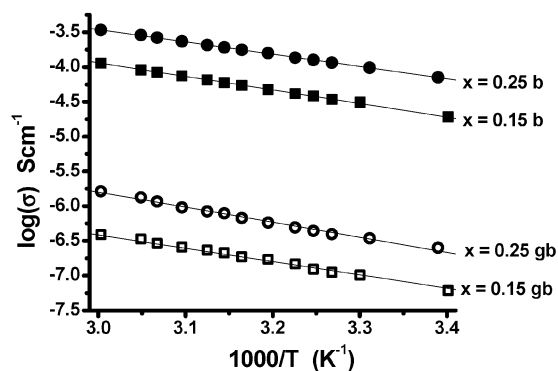


Figure 7. Plots of $\log \sigma_{\text{bulk}}$ and $\log \sigma_{\text{grainboundary}}$ vs. $1000/T(\text{K})$ for $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.15$ and 0.25).

Figure 5a shows the complex impedance diagram at 22 °C. A spike at the lowest frequencies followed by two semicircles depressed below the real axis can be seen. The equivalent circuit which models these results has been represented in Figure 5b. The capacitors have a dependence on frequency of the form $C = B(i\omega)^{-(1-n)}$; these are called “universal capacitors” by Jonscher.¹⁴ The corresponding values of the capacitances are $C_1 = 5 \times 10^{-7}$ F, $C_2 = 1 \times 10^{-7}$ F, and $C_3 = 2 \times 10^{-10}$ F,

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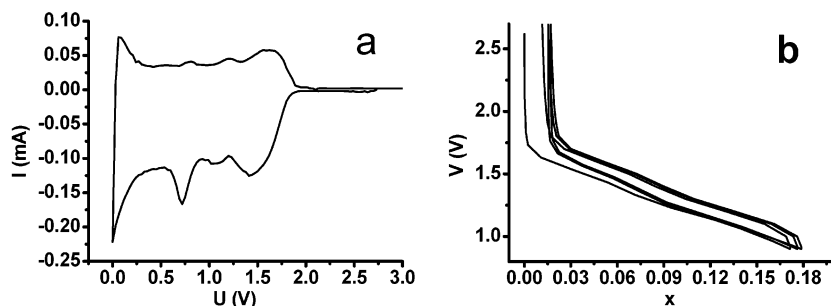


Figure 8. Cyclic voltammogram (a) and voltage–composition curves (b) for Li/1 M LiPF_6 (EC, DMC)/ $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($x = 0.25$).

which are characteristic of the blocking effect of the ionic charge carriers at the electrode–sample interface and the grain boundaries and bulk effects, respectively.¹⁵ The existence of both bulk and grain boundary electrical response is also distinguished in the variation of Z'' and M'' with frequency: two different peaks appear for each of the magnitudes. Figure 6 shows the variation of the real and imaginary parts of the capacitance with the frequency at 22 °C. There is a loss peak accompanied by a nearly saturated $C'(\omega)$ followed by a low-frequency dispersion. This kind of response arises because, at the frequency range $2.5 < \log(f/\text{Hz}) < 4.5$, the behavior is dominated by the resistance of the bulk and the capacitor of the grain boundaries, giving rise to a slope of -1 for C'' (d.c. conductivity, i.e., long-range movement of the Li^+ cations) and a slope of $-1-n$ for C' with values of n for the capacitor close to 1. Such a high value of n is characteristic of a low-loss universal capacitor.¹⁴ This kind of behavior has been observed in systems formed by a bulk material with a high ionic conductivity in series with a low-loss and hence high impedance barrier.¹⁶

Bulk conductivity and grain boundary conductivity values have been determined from the impedance complex diagrams at different temperatures. Figure 7 shows the plots of $\log\sigma_{\text{bulk}}$ and $\log\sigma_{\text{grain boundary}}$ vs $1000/T$ for two compositions. The data fit an Arrhenius-type equation $\sigma = \sigma_0 e^{-E_a/kT}$ where E_a is the activation energy of the process. Table 3 shows the ionic conductivity values at $T = 22$ °C and the activation energies of the conduction process for different oxides of the solid solution. The activation energy values are similar for both the bulk and grain boundary and do not depend on the composition of the material. The conductivity increases with increasing lithium content up to a value corresponding to $x = 0.25$. This behavior is characteristic of a solid solution, in which the increase in the number of charge carriers enhances the conductivity up to a value after which the conductivity decreases due to the

Table 3. Bulk and Grain Boundary Conductivity Values at 300 K and Activation Energies of $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$

x	σ_{bulk} ($\text{S}\cdot\text{cm}^{-1}$)	σ_{gb} ($\text{S}\cdot\text{cm}^{-1}$)	$E_{a, b}$ (eV)	$E_{a, gb}$ (eV)
0.06	1.13×10^{-5}	1.57×10^{-8}	0.37	0.4
0.1	1.37×10^{-5}	3.53×10^{-8}	0.37	0.4
0.15	5.3×10^{-5}	6.10×10^{-8}	0.38	0.4
0.25	7.64×10^{-5}	2.52×10^{-7}	0.35	0.43
0.275	6.04×10^{-5}	1.42×10^{-7}	0.37	0.46
0.3	1.92×10^{-5}	1.57×10^{-8}	0.35	0.4

decrease in the number of nonoccupied sites for the ions to move through.

Electrochemical Behavior of the $\text{La}_{2/3-x}\text{Sr}_x\text{Li}_x\text{TiO}_3$ ($0.06 \leq x \leq 0.3$) System. All the materials of the solid solution have a similar electrochemical behavior. We will again describe those for the sample with composition corresponding to $x = 0.25$. Figure 8a shows a cyclic voltammogram, and Figure 8b shows the voltage–composition curves for three charge and discharge cycles of a cell using the oxide with $x = 0.25$ as working electrode.

Strong cathodic peaks are observed in the voltammogram: the peak at 1.5 V is associated with the reduction of Ti^{4+} to Ti^{3+} due to the electrochemical Li^+ intercalation into the material, and the peak at 0.7 V is ascribed to the decomposition of the electrolyte. In accordance with the potentiostatic experiment is the galvanostatic one, which shows that the material can reversibly insert up to 0.17 mol of lithium atoms per formula unit from 1.75 to 1 V. The very small losses in capacity in the charge processes are most probably associated with the loss of electrical contact between some cathode particles. As a result of the intercalation reaction and therefore, the reduction of Ti^{4+} , electronic conductivity appears in the oxide. This means that the oxide can be used as a solid electrolyte in secondary batteries having Li metal as anode down to 1.75 V.

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